

VertexERP AI — System Architecture Specification

Production Engineering & Multi-Cloud Infrastructure Blueprint

1. Executive Architecture Overview

VertexERP AI is built as a multi-tenant, cloud-native Enterprise Resource Planning and AI Operating System. The platform combines high-throughput transactional ERP capabilities (General Ledger, MES, CRM, Supply Chain, HR) with real-time vector RAG intelligence, MLOps model training, and automated multi-step workflow DAG execution.

2. Cloud Deployment Topology

Global Geo-DNS / Anycast CDN / OWASP WAF DDoS Guard

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US East (AWS EKS Primary Cluster)

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EU Central (AWS EKS Secondary Active)

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APAC (Azure AKS Active Replica)

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API Gateway (FastAPI Uvicorn Async Workers)

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Enterprise React 19 Frontend Web Portal (Vite / Tailwind)

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PostgreSQL 17 Multi-Region Database (PgVector Enabled)

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Redis 7 Cluster (Distributed Session & RBAC Cache)

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FAISS Vector Store (Enterprise Document Embeddings)

3. Platform Module Architecture Matrix

Module Domain	Core Responsibility	Technology Stack
Organization Isolation	Multi-tenant isolation, subsidiaries, cost centers	SQLAlchemy 2 Async, PostgreSQL 17
Identity & Security	OAuth2, JWT bearer, RBAC, ABAC, Security Telemetry	FastAPI Security, Passlib, Cryptography
HR & Payroll Intelligence	Employee lifecycle, time tracking, payroll calculation	Pandas, NumPy, Async Repositories
CRM & Pipeline	Lead tracking, deal forecasting, ticket resolution	Scikit-Learn, LightGBM, SQLAlchemy
Inventory & Logistics	Multi-warehouse stock, serial/lot tracking, valuations	Redis Cache, PostgreSQL B-Tree Indexes
Finance Platform	Double-entry GL, invoicing, accounts payable/receivable	SQLAlchemy Async Transactions, Alembic
Manufacturing MES	Bill of Materials (BOM), Work Orders, OEE Analytics	Async Worker Queue, Manufacturing Engine
Enterprise RAG	Vector search, document chunking, citation retrieval	FAISS, PgVector, SentenceTransformers
AI Copilot Agent	Intent classification, tool execution, NL to SQL	LangChain, Structured Tool Registry
MLOps & ML Studio	AutoML, hyperparameter tuning, model registry	Scikit-learn, XGBoost, SHAP, MLflow

4. Security & Compliance Controls

Enforces strict OWASP Top 10 defenses including Content Security Policy (CSP), HTTP Strict Transport Security (HSTS), AES-256-GCM data encryption at rest, TLS 1.3 in transit, automated HMAC SHA-256 webhook signatures, and immutable SOC 2 audit logging.